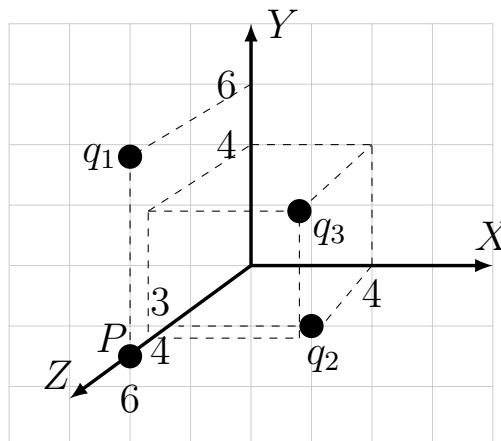


1. (19 points) Se dispone de un sistema formado por 3 cargas puntuales: $q_1 = -12[nC]$, $q_2 = 25[nC]$ y $q_3 = -6[nC]$ ubicadas en los puntos $P_1(0, 6, 6)$, $P_2(4, 0, 3)$ y $P_3(4, 4, 4)$ respectivamente. Considere que las posiciones de las cargas están medidas en centímetros y el valor de $K = 9 \times 10^9 [Nm^2/C^2]$, responda

Nota:

prefijos: $c = 10^{-2}$, $m = 10^{-3}$, $\mu = 10^{-6}$, $n = 10^{-9}$,
 $p = 10^{-12}$.



- (a) (8.0 pts) calcule el potencial eléctrico en un punto $P(0,0,6)$

$$\begin{aligned}
 V_p &= V_{1p} + V_{2p} + V_{3p} \quad ; \quad V_p = \frac{kq_i}{\|r_{ip}\|} \\
 V_p &= \frac{(9 \cdot 10^9)(-12 \cdot 10^{-9})}{(6 \cdot 10^{-2})} + \frac{(9 \cdot 10^9)(25 \cdot 10^{-9})}{(5 \cdot 10^{-2})} + \frac{(9 \cdot 10^9)(-6 \cdot 10^{-9})}{6 \cdot 10^{-2}} \\
 &= \frac{(9)(-12)}{6} \cdot 10^2 + \frac{(9)(25)}{5} \cdot 10^2 + \frac{(9)(-6)}{6} \cdot 10^2 \\
 &= (-18 + 45 - 9) \cdot 10^2 = 18 \cdot 10^2 [V] = 1800 [V] \quad (2,0 \text{ pts})
 \end{aligned}$$

$$\vec{r}_{1p} = \vec{r}_p - \vec{r}_1 = (6\hat{x} - 6\hat{y} - 6\hat{z}) = -6\hat{y} [\mu] \Rightarrow \|\vec{r}_{1p}\| = 6 \cdot 10^{-2} [\mu] \quad (1,0 \text{ pts})$$

$$\vec{r}_{2p} = \vec{r}_p - \vec{r}_2 = (6\hat{x} - 4\hat{x} - 3\hat{z}) = (-4\hat{x} + 3\hat{z}) [\mu] \Rightarrow \|\vec{r}_{2p}\| = 5 \cdot 10^{-2} [\mu] \quad (1,0 \text{ pts})$$

$$\vec{r}_{3p} = \vec{r}_p - \vec{r}_3 = (6\hat{x} - 4\hat{x} - 4\hat{y} - 4\hat{z}) = (-4\hat{x} - 4\hat{y} + 2\hat{z}) [\mu] \Rightarrow \|\vec{r}_{3p}\| = 6 \cdot 10^{-2} [\mu] \quad (1,0 \text{ pts})$$

- (b) (8.0 pts) calcule la energía contenida en el sistema.

$$U_1 = V_0 q_1 = 0 \quad (1)$$

$$U_2 = V_1 q_2 = \frac{kq_1 q_2}{\|\vec{r}_{12}\|} = \frac{(9 \cdot 10^9)(-12 \cdot 10^{-9})(25 \cdot 10^{-9})}{7.81 \cdot 10^{-2}} = \left(\frac{9 \cdot (-12)(25)}{7.81} \right) \cdot 10^{-7} = -345.71 \cdot 10^{-7} = -34.57 [\mu J] \quad (1,0 \text{ pts})$$

$$\begin{aligned}
 U_3 &= (V_1 + V_2) q_3 = \frac{kq_1 q_3}{\|\vec{r}_{13}\|} + \frac{kq_2 q_3}{\|\vec{r}_{23}\|} = \frac{(9 \cdot 10^9)(-12 \cdot 10^{-9})(-6 \cdot 10^{-9})}{4.89 \cdot 10^{-2}} + \frac{(9 \cdot 10^9)(25 \cdot 10^{-9})(-6 \cdot 10^{-9})}{4.12 \cdot 10^{-2}} \\
 &= \frac{(9)(-12)(-6)}{4.89} \cdot 10^{-7} + \frac{(9)(25)(-6)}{4.12} \cdot 10^{-7} = (132.51 - 327.7) \cdot 10^{-7} = -195.19 \cdot 10^{-7} = -19.52 [\mu J] \\
 \therefore U &= U_1 + U_2 + U_3 = -540.9 \cdot 10^{-7} [J] = -54.1 [\mu J] \quad (2,0 \text{ pts})
 \end{aligned}$$

$$\vec{r}_{12} = \vec{r}_2 - \vec{r}_1 = (4\hat{x} + 3\hat{z}) - (6\hat{y} + 6\hat{z}) = (4\hat{x} - 6\hat{y} - 3\hat{z}) [\mu] \Rightarrow \|\vec{r}_{12}\| = \sqrt{61} [\mu] = 7.81 \cdot 10^{-2} [\mu] \quad (1,0 \text{ pts})$$

$$\vec{r}_{13} = \vec{r}_3 - \vec{r}_1 = (4\hat{x} + 4\hat{y} + 4\hat{z}) - (6\hat{y} + 6\hat{z}) = (4\hat{x} - 2\hat{y} - 2\hat{z}) [\mu] \Rightarrow \|\vec{r}_{13}\| = \sqrt{24} = 2\sqrt{6} \cdot 10^{-2} = 4.89 \cdot 10^{-2} [\mu] \quad (1,0 \text{ pts})$$

$$\vec{r}_{23} = \vec{r}_3 - \vec{r}_2 = (4\hat{x} + 4\hat{y} + 4\hat{z}) - (4\hat{x} + 3\hat{z}) = (4\hat{y} + \hat{z}) [\mu] \Rightarrow \|\vec{r}_{23}\| = \sqrt{17} = 4.12 \cdot 10^{-2} [\mu] \quad (1,0 \text{ pts})$$

- (c) (3.0 pts) calcule el trabajo necesario en traer una carga $q_4 = 14[nC]$ al punto P.

$$\begin{aligned}
 W &= U = q_4 V_p \\
 &= 14 \cdot 10^{-9} (18 \cdot 10^2) \quad (1,0 \text{ pts}) \\
 &= 252 \cdot 10^{-7} [J] \\
 &= 25.2 \cdot 10^{-6} [J] \quad (2,0 \text{ pts})
 \end{aligned}$$